PSO201/PHY204: Mid-Sem Exam

Date: Feb 28, 2025; Maximum Marks: 60

Answer all questions belonging to each group together in separate places.

GROUP A

1. Answer the following short answer type questions (10×2=20)

a) An electron is moving in a periodic potential $V(x) = V(x + na)$, where n is any integer. The probability of finding the electron at $x = a$ is C, where C is a real number. What is the probability of finding the electron at $x = na$?

$$\psi(x+na) = e^{ikna}\,\psi(x)$$

$$|\psi(x+na)|^2 = |\psi(x)|^2 = c$$

b) Identify the correct (put a √) and the false (put a ×) statements:

i) If a solution to the Schrodinger equation in presence of potential $V(x)$ has the property $\psi(x) \to 0$, as $x \to \pm\infty$, then the solution must be non-degenerate and real, apart from a possible overall phase factor. ✓

ii) For a stationary state wavefunction in 3 dimensional space, the probability current desnity $\vec{J}(\vec{x}, t)$ is divergence free, i.e. $\vec{\nabla}.\vec{J} = 0$. ✓

iii) Superposition of two stationary states of different energies is also stationary. ×

iv) The Bloch wavefunction $\psi_k(x) = e^{ikx}u_k(x)$ which describes the quantum state of an electron in a periodic potential is a momentum eigenstate. ×

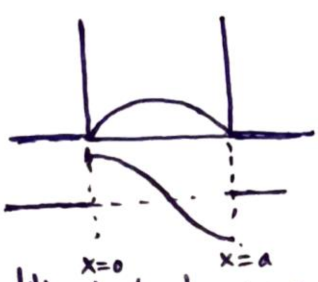
c) For a double delta potential $V(x) = -V_0\,\delta(x + a) - V_0\,\delta(x - a)$, with $V_0 > 0$ such that there are two bound eigenstate solutions, what is the probability of finding a particle at $x = 0$ in the first excited bound eigenstate. There is no need to work out the solution! Use symmetry arguments to find your answer.

The potential is symmetric w.r.t $x=0$

The eigenfunction corresponding to 1st excited state will be odd.

The probability of finding the particle at $x=0$ is zero.

d) Plot the ground state wavefunction $\psi(x)$ and the corresponding $\frac{d\psi(x)}{dx}$ for all x for an infinite square well potential of width a. Comment on the origin of (dis)continuity of $\frac{d\psi(x)}{dx}$. Keep your answer brief (one or two sentences only).



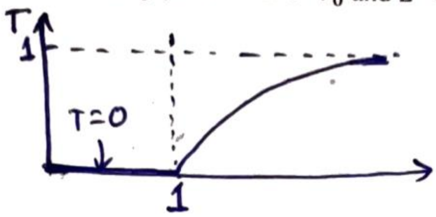
$$\lim_{\epsilon\to 0} \left.\frac{d\psi}{dx}\right|_{a+\epsilon} - \left.\frac{d\psi}{dx}\right|_{a-\epsilon} = \lim_{\epsilon\to 0} \frac{2m}{\hbar^2}\int_{a-\epsilon}^{a+\epsilon} [V(x)-E]\,\psi(x)\,dx$$

$V(x) = \infty$ at $x=a$ in the limit $\epsilon\to 0$

$$\Rightarrow \left.\frac{d\psi}{dx}\right|_{a+} - \left.\frac{d\psi}{dx}\right|_{a-} \neq 0$$

$\frac{d\psi}{dx}$ discontinuous at $x = 0, a$

e) A particle of mass m and energy E is moving from left to right and incident on a potential step of height V_0, at $x = 0$ and extending along $x > 0$. Sketch the transmission probability T as a function of E/V_0 (for both $E > V_0$ and $E < V_0$).



f) Consider a particle moving on a circle $x \in [0, L]$. Is the energy of the particle positive or negative? Justify your answer.

$$-\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2} = E\psi \Rightarrow -\frac{\hbar^2}{2m}\int_0^L \psi^* \frac{d}{dx}\frac{d\psi}{dx}\,dx = E\int_0^L \psi^*\psi\,dx = E$$

$$\Rightarrow -\frac{\hbar^2}{2m}\int_0^L dx\left[\frac{d}{dx}\left(\psi^*\frac{d\psi}{dx}\right) - \frac{d\psi^*}{dx}\frac{d\psi}{dx}\right] = E$$

$$\Rightarrow -\frac{\hbar^2}{2m}\left(\underbrace{\left[\psi^*\frac{d\psi}{dx}\right]_0^L}_{0 \text{ since } \psi^*,\ \frac{d\psi}{dx} \text{ periodic in } L} - \int_0^L dx\left|\frac{d\psi}{dx}\right|^2\right) = E \quad \Rightarrow E = \frac{\hbar^2}{2m}\underbrace{\int_0^L dx\left|\frac{d\psi}{dx}\right|^2}_{+ve} > 0$$

g) Identify the correct (put a √) and the false (put a ×) statements:
A) The eigenvalues of a Hermitian operator are, in general, complex. ✗
B) If two operators commute with each other they will have a complete set of common eigenfunctions. ✓
C) The normalized non-degenerate eigenfunctions of a Hermitian operator are orthogonal to each other. ✓
D) If ψ is an eigenstate of an observable $\hat{Q}$, the uncertainty ΔQ in the state ψ is zero. ✓

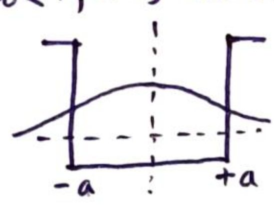
h) If an observable $\hat{Q}$ commutes with the Hamiltonian, find $\frac{d\langle\hat{Q}\rangle}{dt}$ for any state ψ. Justify your answer (No derivation required).

$$i\hbar\frac{d}{dt}\langle\hat{Q}\rangle_\psi = \langle[\hat{Q},\hat{H}]\rangle_\psi = 0$$

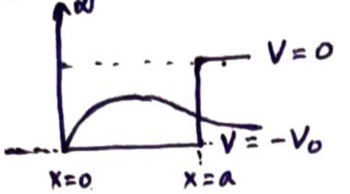
$$\Rightarrow \frac{d\langle\hat{Q}\rangle_\psi}{dt} = 0$$

i) Consider a finite square well potential of depth V_0 and width $2a$, i.e. $V(x) = -V_0$ for $|x| \leq a$ and $V(x) = 0$ elsewhere; where V_0 and a are real, positive constants. If $z_0{}^2 = \frac{2mV_0a^2}{\hbar^2}$, what is the number of bound states if $z_0 < 1$. Sketch the ground state wave function.

For $z_0 < 1$, only one bound state → even solution with no nodes.



j) Consider a one-dimensional square-well potential $V(x) = \infty$, for $x < 0$; $V(x) = -V_0$, for $0 < x < a$; $V(x) = 0$, for $x > a$. Sketch the ground state wavefunction using physical reasoning.



$\psi(0) = 0$ since $V(0) = \infty$
Finite step potential at $x = a$
$\Rightarrow$ leakage of wave function for $x > a$, $\psi(\infty) = 0$
$\frac{d\psi}{dx}$ continuous at $x = a$ and discontinuous at $x = 0$

GROUP B

2. a) Consider a particle of mass m on a circle $x \in [0, L]$.
   i) Find out the energy and momentum in the nth eigenstate $\psi_n(x)$.
   ii) Find out the ground state energy and momentum.
   iii) Identify the degenerate eigenstates for $n \neq 0$.
   iv) Construct a linear superposition of the degenerate eigenstates such that the resulting energy eigenstate is not a momentum eigenstate.
   v) Write down the expression for the normalized time dependent wavefunction $\psi_n(x, t)$.
   [2+1+1+2+2]

a) $\frac{d^2\psi}{dx^2} = -\left(\frac{2mE}{\hbar^2}\right)\psi = -k^2\psi \quad k^2 = \frac{2mE}{\hbar^2}, \quad E = \frac{\hbar^2 k^2}{2m}$

$\psi \sim e^{ikx}$

i) $e^{ik(x+L)} = e^{ikx} \Rightarrow e^{ikL} = 1, \quad kL = 2n\pi, \quad n = 0, \pm1, \pm2 \ldots$

$P_n = \hbar k_n = \frac{2\pi\hbar n}{L}$

$E_n = \frac{\hbar^2 k_n^2}{2m} = \frac{2\pi^2\hbar^2 n^2}{mL^2}$

ii) ground state : $n = 0 \Rightarrow$ state with zero energy and momentum

iii) $n = \pm1$ corresponds to degenerate eigenstates with momentum $\pm\frac{2\pi\hbar}{L}$
similarly $n = \pm2, \pm3, \ldots$ etc are degenerate eigenstates with opposite momenta $\pm k_n$

iv) $\psi_{\pm k} \rightarrow$ degenerate eigenstates $\Rightarrow \psi_{+k} = e^{ikx}, \psi_{-k} = e^{-ikx}$

$\psi_k \pm \psi_{-k}$ are also energy eigenstates

$\left.\begin{array}{l}\psi_k + \psi_{-k} \sim \cos kx \\ \psi_k - \psi_{-k} \sim \sin kx\end{array}\right\}$ not momentum eigenstates (Apply $\frac{\hbar}{i}\frac{\partial}{\partial x}$ on $\cos kx$ or $\sin kx$)

v) $\psi_n(x) = c\, e^{ik_n x} \Rightarrow \int_0^L dx\, |\psi_n|^2 = 1 \Rightarrow c = \frac{1}{\sqrt{L}}$

$\psi_n(x,t) = \psi_n(x)\, e^{-iE_n t/\hbar} = \frac{1}{\sqrt{L}} e^{ik_n x} e^{-iE_n t/\hbar}$, where $k_n = \frac{2\pi n}{L}$
and $E_n = \frac{2\pi^2\hbar^2 n^2}{mL^2}$

b) Suppose that the wavefunction for a particle on a circle $x \in [0, L]$ is given by $\psi(x) = \sqrt{\frac{2}{L}}\left(\frac{1}{\sqrt{3}}\sin\frac{2\pi x}{L} + \sqrt{\frac{2}{3}}\cos\frac{6\pi x}{L}\right)$. If you measure the momentum in this state, what are the possible values of momentum and the corresponding probabilities? Find out the expectation value of the momentum in the state $\psi(x)$. [6+2]

$$\psi(x) = \sqrt{\frac{2}{L}}\left(\frac{1}{\sqrt{3}}\sin\frac{2\pi x}{L} + \sqrt{\frac{2}{3}}\cos\frac{6\pi x}{L}\right)$$

$$= \frac{\sqrt{2}}{\sqrt{L}}\left[\frac{1}{\sqrt{3}}\frac{1}{2i}\left(e^{2\pi i x/L} - e^{-2\pi i x/L}\right) + \sqrt{\frac{2}{3}}\frac{1}{2}\left(e^{6\pi i x/L} + e^{-6\pi i x/L}\right)\right]$$

The momentum eigenstates: $\psi_n(x) = \frac{1}{\sqrt{L}} e^{2\pi i n x/L}$ $\quad n = 0, \pm1, \pm2, \ldots$

$$\psi(x) = \sqrt{\frac{2}{3}}\frac{1}{2i}\psi_1(x) - \sqrt{\frac{2}{3}}\frac{1}{2i}\psi_{-1}(x) + \frac{1}{\sqrt{3}}\psi_3 + \frac{1}{\sqrt{3}}\psi_{-3}$$

Momentum eigenvalues $P_n = \frac{2\pi\hbar n}{L}$

In this case, the possible values of momentum: $\pm\frac{2\pi\hbar}{L}$, $\pm\frac{6\pi\hbar}{L}$

probability for $P_1 = +\frac{2\pi\hbar}{L}$: $\left|\sqrt{\frac{2}{3}}\frac{1}{2i}\right|^2 = \frac{1}{6}$

$P_{-1} = -\frac{2\pi\hbar}{L}$: $\frac{1}{6}$

$P_3 = +\frac{6\pi\hbar}{L}$: $\left|\frac{1}{\sqrt{3}}\right|^2 = \frac{1}{3}$

$P_{-3} = -\frac{6\pi\hbar}{L}$: $\frac{1}{3}$

Expectation value of momentum

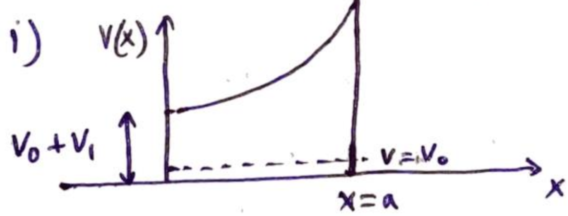
$$= \frac{1}{6}\cdot\frac{2\pi\hbar}{L} + \frac{1}{6}\left(-\frac{2\pi\hbar}{L}\right) + \frac{1}{3}\left(\frac{6\pi\hbar}{L}\right) + \frac{1}{3}\left(-\frac{6\pi\hbar}{L}\right) = 0$$

c) A particle of mass m and energy V_0 moves from left to right (along positive x-direction) in a potential given by $V(x) = V_0 + V_1 e^{2\beta x}$ ($V_0 \ll V_1$) for $0 < x < a$ and zero elsewhere.

i) Sketch the potential $V(x)$.

ii) Estimate the probability that the particle will be found in the region $x > a$. [1+3]

i)



At $x = 0$, $V(x) = V_0 + V_1$

$x = a$, $V(x) = V_0 + V_1 e^{2\beta a}$

Since $V_0 \lll V_1$, $T = e^{-2\int K(x)dx}$

$$K(x) = \sqrt{\frac{2m}{\hbar^2}(V(x) - E)} = \sqrt{\frac{2m}{\hbar^2}\left[V_0 + V_1 e^{2\beta x} - V_0\right]} = \sqrt{\frac{2mV_1}{\hbar^2}}\, e^{\beta x}$$

$$\Rightarrow \int_0^L K(x)\,dx = \sqrt{\frac{2mV_1}{\hbar^2}}\int_0^a e^{\beta x}dx = \sqrt{\frac{2mV_1}{\hbar^2}}\frac{1}{\beta}\left(e^{\beta a} - 1\right)$$

$$\Rightarrow T = e^{-\left\{\frac{2}{\beta}\sqrt{\frac{2mV_1}{\hbar^2}}\left(e^{\beta a} - 1\right)\right\}}$$

GROUP C

3. a) The wavefunction of a particle in position space is given by, $\psi(x) = \frac{2}{a^{3/2}} x e^{-x/a}$ for $x > 0$ and zero elsewhere.

i) Find the most probable value of position of the particle. [2]

ii) Find the corresponding wavefunction in momentum space. [5]

iii) Find the probability density in momentum space. [2]

iv) Find the most probable value of the momentum of the particle. [1]

$$\psi(x) = \frac{2}{a^{3/2}} x e^{-x/a} \quad , x>0$$
$$= 0 \quad , x \leq 0$$

i) Most probable value of position corresponds to the peak position of $|\psi(x)|^2$

since $x>0$, $\psi(x)$ is positive, $\psi(x)$ is real

$$\frac{d\psi}{dx} = \frac{2}{a^{3/2}}\left(1-\frac{x}{a}\right) e^{-x/a} \Rightarrow \frac{d\psi}{dx} = 0 \text{ at } x=a$$

$\Rightarrow x=a$ is the ~~po~~ most probable value of position

ii) $$\Phi(p) = \frac{1}{\sqrt{2\pi\hbar}} \int_{-\infty}^{\infty} \psi(x) e^{-ipx/\hbar} dx = \frac{1}{\sqrt{2\pi\hbar}} \frac{2}{a^{3/2}} \int_0^{\infty} x e^{-x/a} e^{-ipx/\hbar} dx$$

$$= \frac{1}{\sqrt{2\pi\hbar}} \frac{2}{a^{3/2}} \int_0^{\infty} x e^{-\left(\frac{1}{a}+\frac{i}{\hbar}p\right)x} dx$$

Integrate by parts, $$\Phi(p) = \frac{1}{\sqrt{2\pi\hbar}} \frac{2}{a^{3/2}} \frac{(-1)}{\left(\frac{1}{a}+\frac{i}{\hbar}p\right)} \int_0^{\infty} e^{-\left(\frac{1}{a}+\frac{i}{\hbar}p\right)x} dx$$

$$= \frac{1}{\sqrt{2\pi\hbar}} \frac{2}{a^{3/2}} \frac{1}{\left(\frac{1}{a}+\frac{i}{\hbar}p\right)^2} = \sqrt{\frac{2\hbar^3 a}{\pi}} \frac{1}{(ipa+\hbar)^2}$$

iii) $$|\Phi(p)|^2 = \frac{2\hbar^3 a}{\pi} \frac{1}{(p^2a^2+\hbar^2)^2}$$

iv) $|\Phi(p)|^2$ is maximum for $p=0 \Rightarrow$ Most ~~p~~ probable value of momentum is $p=0$

b) Consider the scattering of particle off an attractive delta function potential of strength V_0 located at the origin, $V(x) = -V_0\,\delta(x)$, with $V_0 > 0$. A particle of mass m is incident from the left with energy E. Calculate the transmission probability T. Write your answer in terms of E, m, V_0 and other fundamental constant(s). What should be the ratio of the energy of the particle E and the bound state energy of the delta potential $|E_b|$ such that the transmission probability is $\frac{1}{2}$? [8+2]

$$\psi(x) = Ae^{ikx} + Be^{-ikx}, \quad x<0$$
$$= Ce^{ikx}, \quad x>0 \qquad k = \sqrt{\frac{2mE}{\hbar^2}}$$

$\psi(x)$ continuous at $x=0 \Rightarrow A+B=C$ --- (1)

$\frac{d\psi}{dx}$ discontinuous at $x=0 \Rightarrow \left.\frac{d\psi}{dx}\right|_{0+} - \left.\frac{d\psi}{dx}\right|_{0-} = -\frac{2mV_0}{\hbar^2}\psi(0)$

$$\Rightarrow ikC - ik(A-B) = -\frac{2mV_0}{\hbar^2}C$$

$$\Rightarrow B-A = -\left(\frac{2mV_0}{ik\hbar^2}+1\right)C \quad \text{--- (2)}$$

(1)-(2) $\Rightarrow \quad 2A = \left(\frac{2mV_0}{ik\hbar^2}+2\right)C \Rightarrow \frac{C}{A} = \frac{1}{1-\frac{imV_0}{k\hbar^2}}$

$J_{ref} = \frac{\hbar k}{m}|B|^2$, $J_{inc} = \frac{\hbar k}{m}|A|^2$, $J_{trans} = \frac{\hbar k}{m}|C|^2$

$$T = \frac{J_{trans}}{J_{inc}} = \left|\frac{C}{A}\right|^2 = \frac{1}{1+\frac{mV_0^2}{2\hbar^2 E}} \qquad \left(\text{using } k^2 = \frac{2mE}{\hbar^2}\right)$$

Again bound state energy $E_b = -\frac{mV_0^2}{2\hbar^2} \Rightarrow |E_b| = \frac{mV_0^2}{2\hbar^2}$

$$\Rightarrow T = \frac{1}{1+\frac{|E_b|}{E}} \Rightarrow \text{when } E = |E_b|,\ T = \frac{1}{2}$$